Project Name: Football Diplomacy

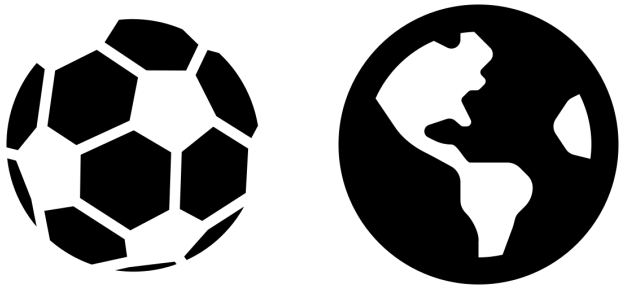
GitHub

Project Description: Do Geopolitics Predict Where Football Talent Flows and Who Plays Whom?

A country-pair × year machine-learning study of player transfers and international friendlies

Submitted to: Dr Shweta Meena

Subject: AI for Sports (SE-333)



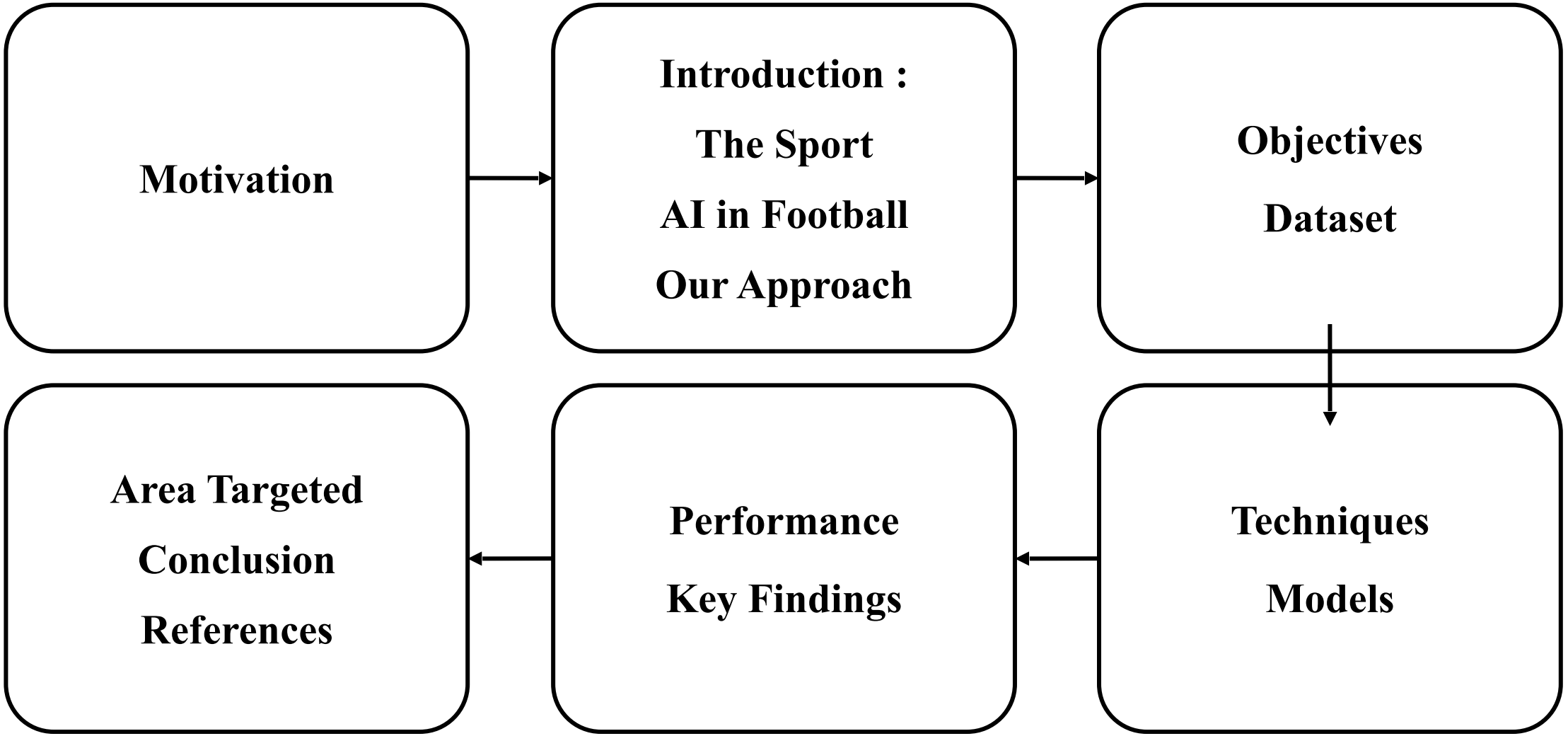
Sports Analytics

Predicting international transfers and fixtures, and measuring their political drivers



Name: Pranit Madan [Roll No: 24/SE/128]

Outline

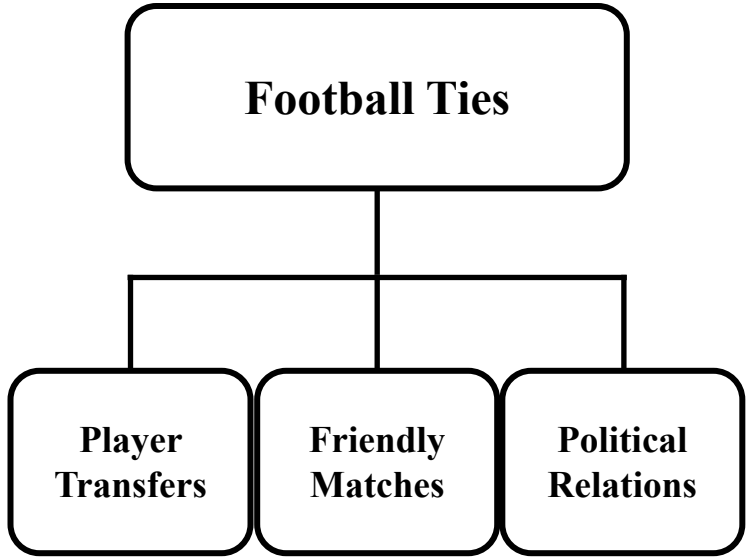

Motivation
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Conclusion
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Motivation

"Does Politics Shape the Beautiful Game?"

Understanding the Problem

- Football is the most global sport: FIFA has 211 member associations, more than the UN's 193 members
- Every year clubs sign thousands of players across borders, and national teams choose whom to play in friendlies
- Leaders often claim "football diplomacy" builds bridges, but the evidence is mostly anecdotes
- No large-scale, data-driven test of whether politics predicts football ties between countries

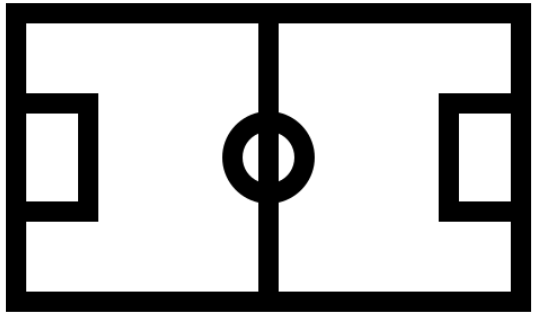


Introduction

The Sport: Association Football

- 11-a-side game governed by FIFA and six confederations: UEFA, CONMEBOL, CONCACAF, CAF, AFC, OFC
- Countries are linked in two ways: club transfers (market decisions) and national-team matches (federation decisions)
- Competitive matches are fixed by tournament draws; friendlies are chosen freely, so they reveal preferences
- The transfer market moves talent from many origin countries into a few rich leagues

"Friendlies are chosen; competitive matches are scheduled."



How AI Is Already Integrated in Football

Player Tracking & Performance

Optical/GPS tracking, expected goals (xG), physical load

Recruitment & Scouting

Data-driven player search, similarity and valuation models

Officiating

VAR and semi-automated offside technology (FIFA World Cup 2022)

Injury Risk

Wearables + ML to flag overload and injury risk

Tactics & Opponent Analysis

Video analysis, pass networks, set-piece design

Fan Engagement

Personalised content, automated highlights, chatbots

Gap: AI in football looks **inside the game**; almost none looks at football **between countries**.

Introduction

Problem Statement

- Football ties between countries are very uneven: most country pairs never exchange a player or play a friendly
- Gravity factors (distance, economic size, shared language) probably explain much of the pattern
- Unknown: does geopolitics (UN voting alignment, colonial history, armed conflict) add anything beyond gravity?
- Does political tension reduce football ties, and does football bring countries closer?

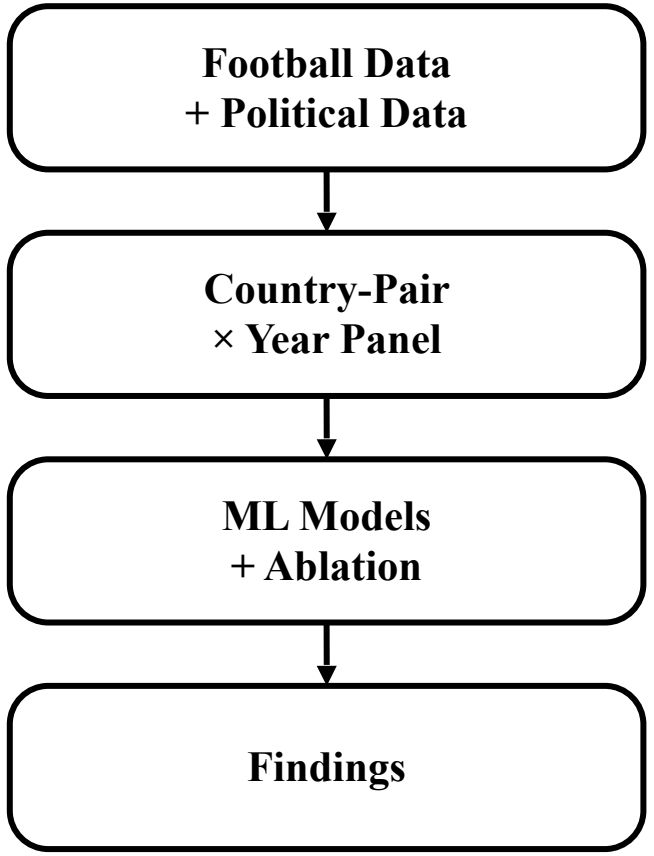
"Do Geopolitics Predict Where Football Talent Flows and Who Plays Whom?"



Introduction

Our Approach

- Built a country-pair × year panel joining football data with geopolitical data
- Two targets: T1 players moving from a country to a league country (2010–21); T2 friendlies between two countries (1994–2021)
- Ablation: Gravity (G) vs Gravity + Geopolitics (G+P) vs + Conflict (G+P+C), each with and without last year's value
- Count models: Poisson GLM, Random Forest, HistGradientBoosting, Two-Part hurdle model
- Robustness: within-pair test, placebo test, natural experiments (Qatar, Russia 2022)



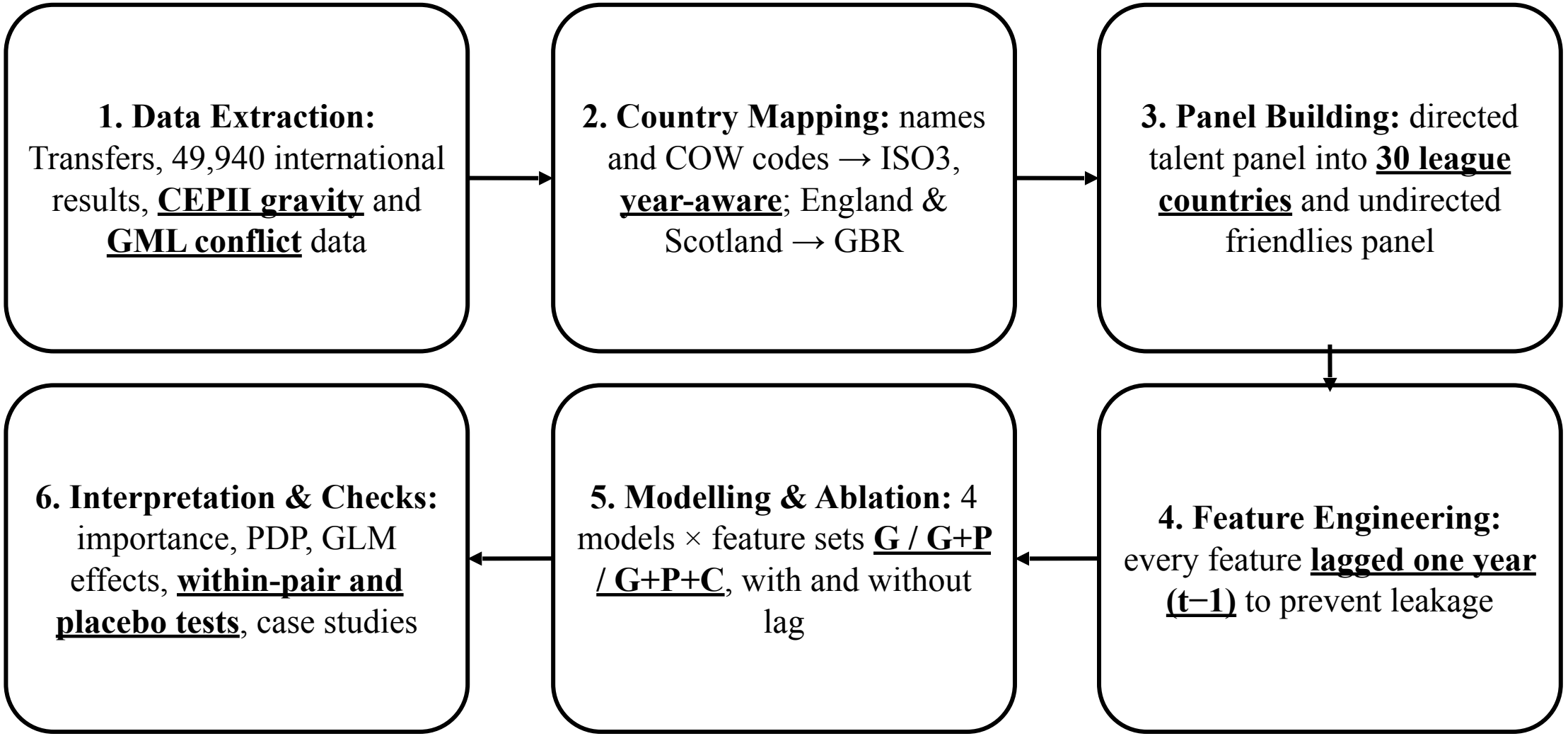
Objectives

1. **Predict:** cross-border player flows and friendlies on years and country pairs the model has never seen
2. **Measure:** how much geopolitics adds beyond gravity (ablation)
3. **Explain:** the direction and size of each factor with interpretable effects
4. **Test:** whether politics changes football within the same country pair over time
5. **Reverse:** check whether football predicts countries moving closer politically

Dataset

S.No.	Dataset	What We Used	Source (Webpage URL)
1	Football Data from Transfermarkt (D. Cariboo)	175,165 transfers, 50,149 players, clubs and leagues	github.com/dcaribou/transfermarkt-datasets
2	International Football Results 1872–2026	49,940 men's international matches	github.com/martj42/international_results
3	CEPII Gravity Database V202211	Distance, GDP, population, language, colonial ties, UN voting disagreement	cepii.fr (Gravity database)
4	GML Militarized Interstate Disputes 2.2.1	Dispute onset, ongoing dispute, hostility level (to 2010)	svmiller.com/gml-mid-data
5	UN General Assembly Resolution ES-11/1 votes	193 member votes on Russia–Ukraine (2 Mar 2022)	Wikipedia: Resolution ES-11/1

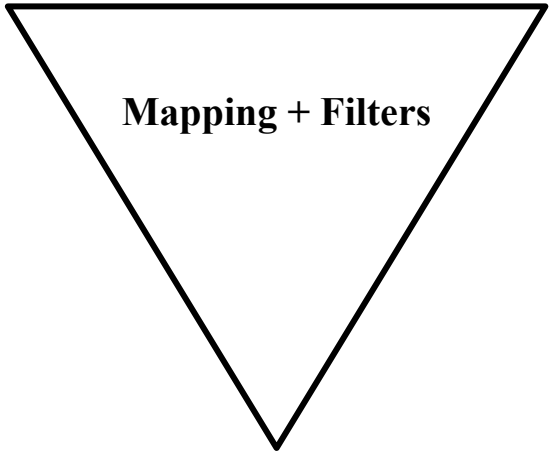
Methodology



Techniques - Data Preprocessing

- **Country harmonisation:** year-aware mapping to ISO3; USSR → RUS, Yugoslavia → SRB, West Germany → DEU; Czechoslovakia and East Germany excluded, never split
- **Destination universe:** club → league → country gives 30 league countries (England + Scotland = GBR)
- **Coverage check:** cross-border transfer records are sparse before 2010, so the T1 window is 2010–2021
- **Zero-filling:** only pairs where both countries existed that year; domestic moves removed

175,165 Raw Transfers



16,071 Cross-Border Moves
into 30 leagues, 2010–2021

49,940 Matches → 8,399 Friendlies
across 823,284 pair-years (1994–2021)

Techniques - Feature Engineering

- **Gravity (G):** log distance, contiguity, common language, log GDP and population, trade agreements, EU, WTO, market size (total inflow / outflow)
- **Geopolitics (P):** UN voting disagreement, colonial relationship (post-1945 and ever), common coloniser
- **Conflict (C):** militarised dispute onset, ongoing dispute, highest hostility level
- **Lag (L):** last year's flow, plus the reverse flow for transfers
- **Leakage control:** every feature is measured in year t−1 for target year t

[Gravity] = 72.6% Deviance Explained (T1)

[Gravity] + [Geopolitics] = 72.4%

Techniques - Model Training and Testing

- **Time split:** train on years ≤ 2016, test on 2017–2019; 2020–21 reported separately as a COVID stress test
- **GroupKFold (5 folds):** whole country pairs held out (A→B and B→A together), so the model is tested on unseen pairs
- **Pipeline security:** imputation and scaling fitted inside training folds only; random_state = 42
- **Overfitting check:** train vs test gap; best T1 model gap = 0.04



GroupKFold by Country Pair

Methodology: Models Compared

Poisson GLM

Linear baseline for counts. Gives **interpretable incidence-rate ratios**.

Random Forest

Parallel ensemble of 150 trees; captures **non-linear interactions**.

HistGradientBoosting (Poisson loss)

Sequential boosting; each tree corrects the previous one; **handles missing values**.

Two-Part Hurdle (Champion)

Step 1: will any link exist? Step 2: how many? **P(link) × E(count)**.

Methodology: Two-Part Model (Champion)

Type & Validation:

Hurdle model from two gradient-boosted learners. Validated on the time split and GroupKFold.

Purpose:

About **93% of talent pair-years and 99% of friendly pair-years are zero**; handle the zeros explicitly.

Mechanism:

A classifier estimates **P(link > 0)**; a Poisson booster estimates the size given a link; the two are multiplied.

Empirical Outcome:

Best for both targets: T1 **72.4% deviance explained, ROC-AUC 0.93**; T2 **26.5%, ROC-AUC 0.90**.

Techniques - Evaluation Metrics

1. Poisson Deviance = (2/n) Σ [y·ln(y/ŷ) − (y − ŷ)]

Error measure designed for counts (lower is better).

2. D^2 (Deviance Explained) = 1 − Dev(model) / Dev(mean-only model)

The R^2 of count data: share of the variation the model explains.

3. ROC-AUC and PR-AUC

ROC-AUC: chance a linked pair is ranked above an unlinked pair. PR-AUC focuses on the rare links.

4. MAE = (1/n) Σ |y − ŷ| and Spearman ρ (ranking)

Average absolute error, and how well pairs are ordered.

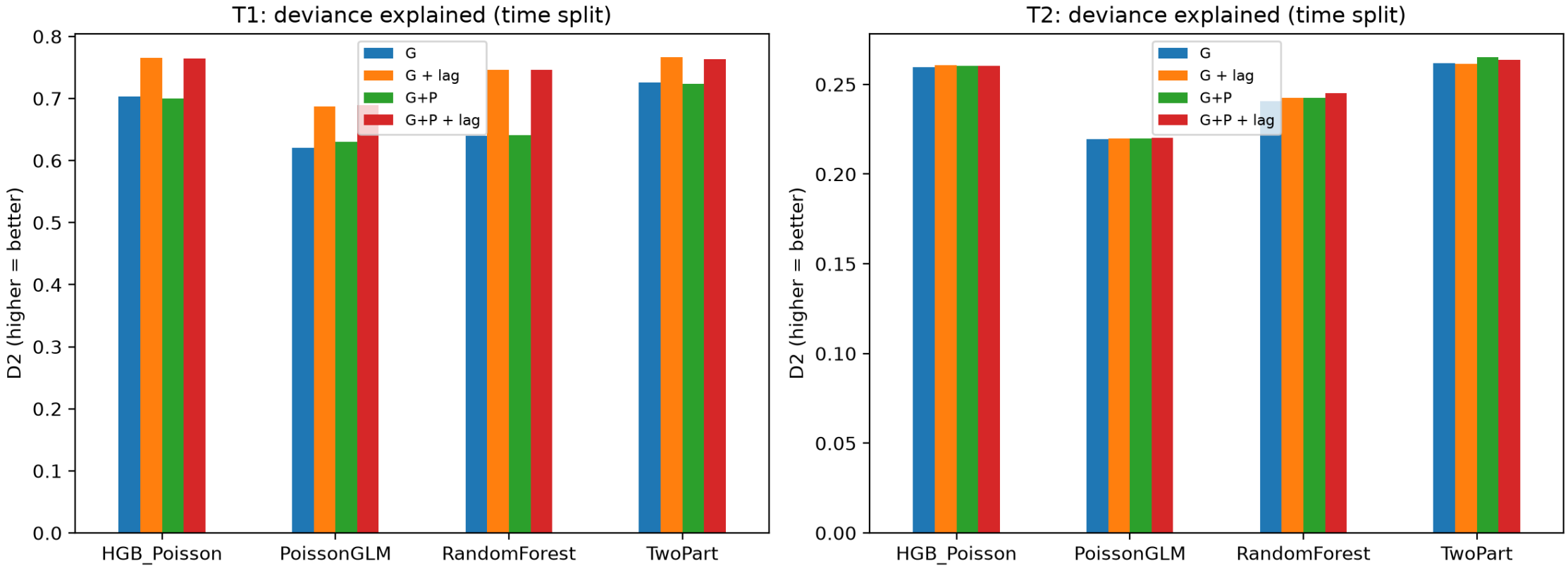
Why not accuracy? 99% of friendly pair-years are zero, so always predicting "no match" would score **99% while learning nothing**.

Performance

Target / Model	D^2 Time Split (no lag)	D^2 Time Split (+ lag)	D^2 Unseen Pairs (no lag)	ROC-AUC (time split)	Gain from Geopolitics (ΔD^2)
T1 Talent – Two-Part (Best)	72.4%	76.4%	64.2%	**0.93**	−0.2 / +0.4 pts
T1 Talent – Persistence (last year)	n/a*	n/a*	n/a*	0.79	—
T2 Friendlies – Two-Part (Best)	26.5%	26.4%	32.5%	**0.90**	+0.3 / +0.5 pts
T2 Friendlies – Persistence (last year)	n/a*	n/a*	n/a*	0.53	—

**Persistence predicts exactly 0 wherever last year was 0, so its deviance-based score is undefined; it is judged on ranking (ROC-AUC). Gain = time split / unseen pairs.*

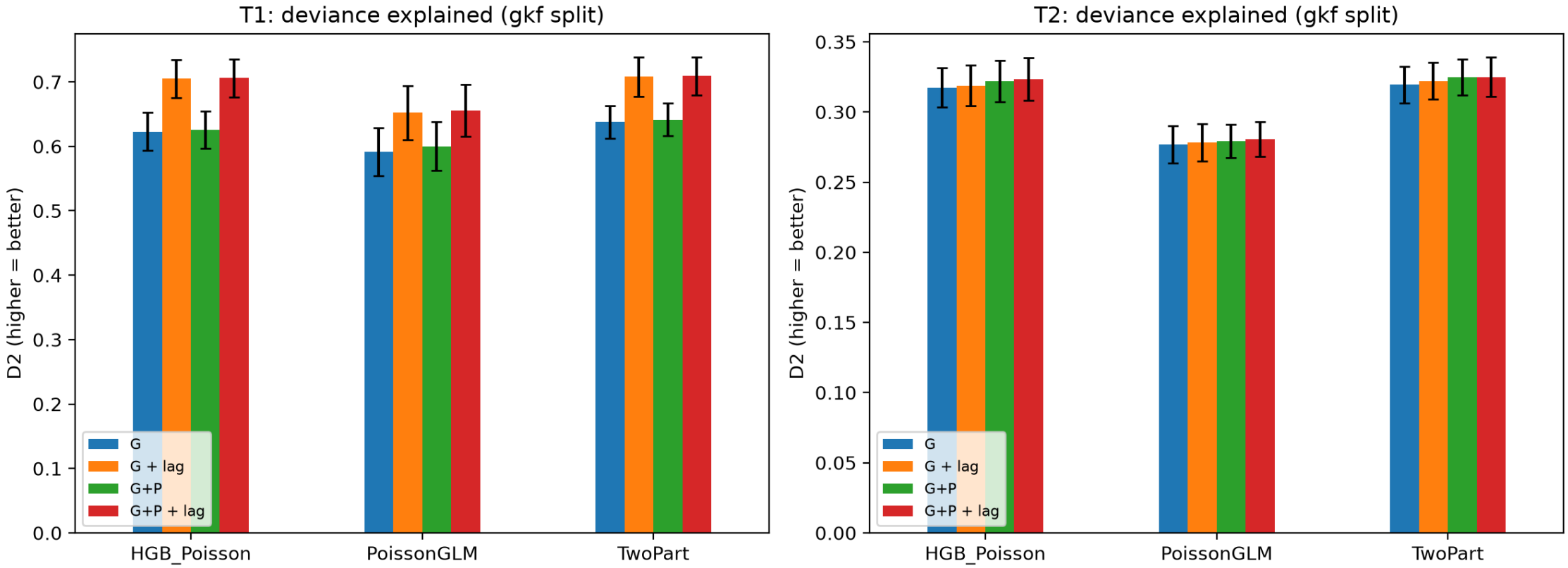
Performance



Ablation: Gravity (G) vs Gravity + Geopolitics (G+P)

Blue (G) and green (G+P) bars are almost the same height: geopolitics adds < 0.5 points.

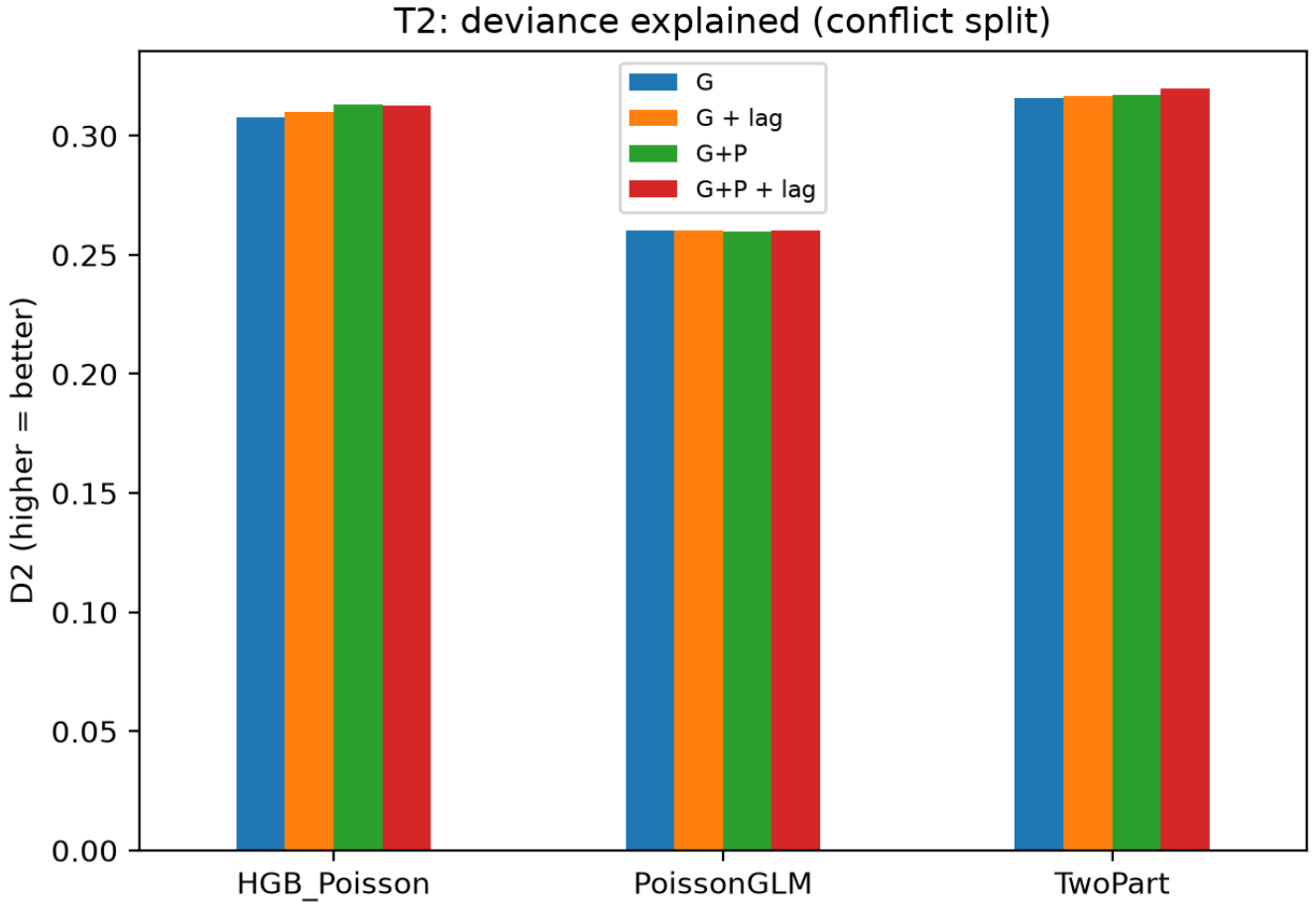
Performance



Unseen Country Pairs (GroupKFold, mean ± std)

Results hold on country pairs the model never saw, so it is not memorising pairs.

Performance: Does Conflict Add Anything?



Conflict Window

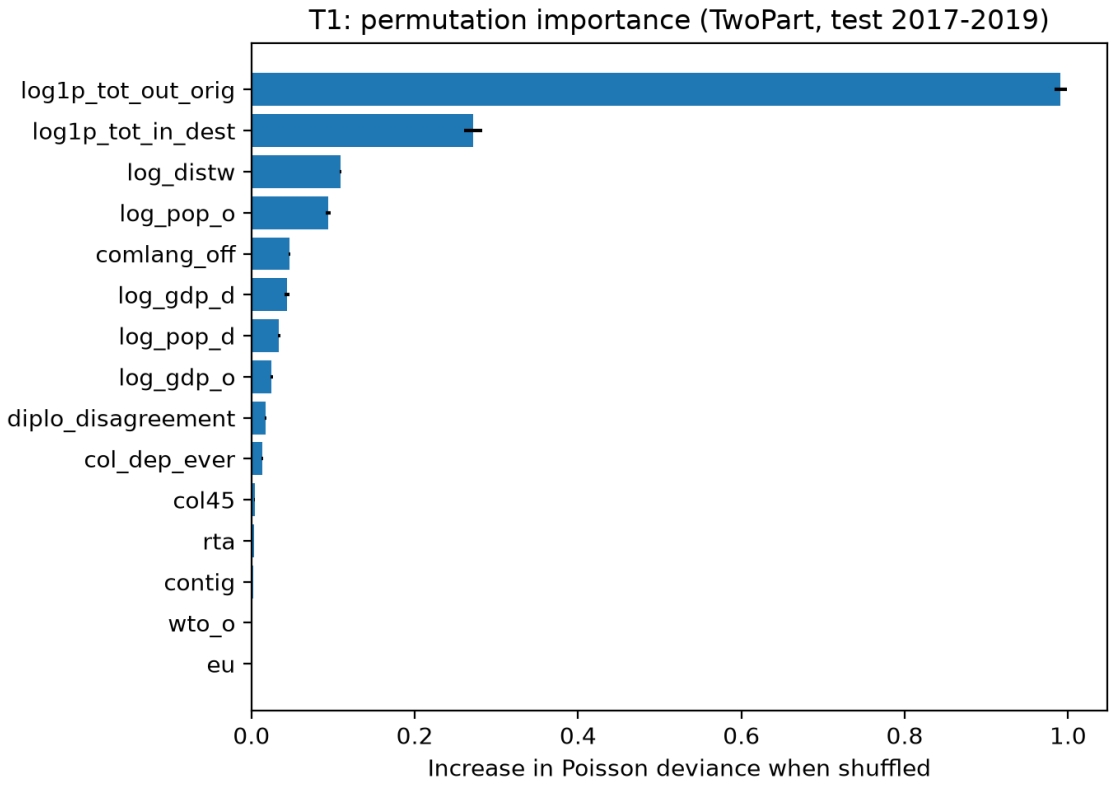
Friendlies, target years 1994–2011 (GML conflict data ends in 2010)

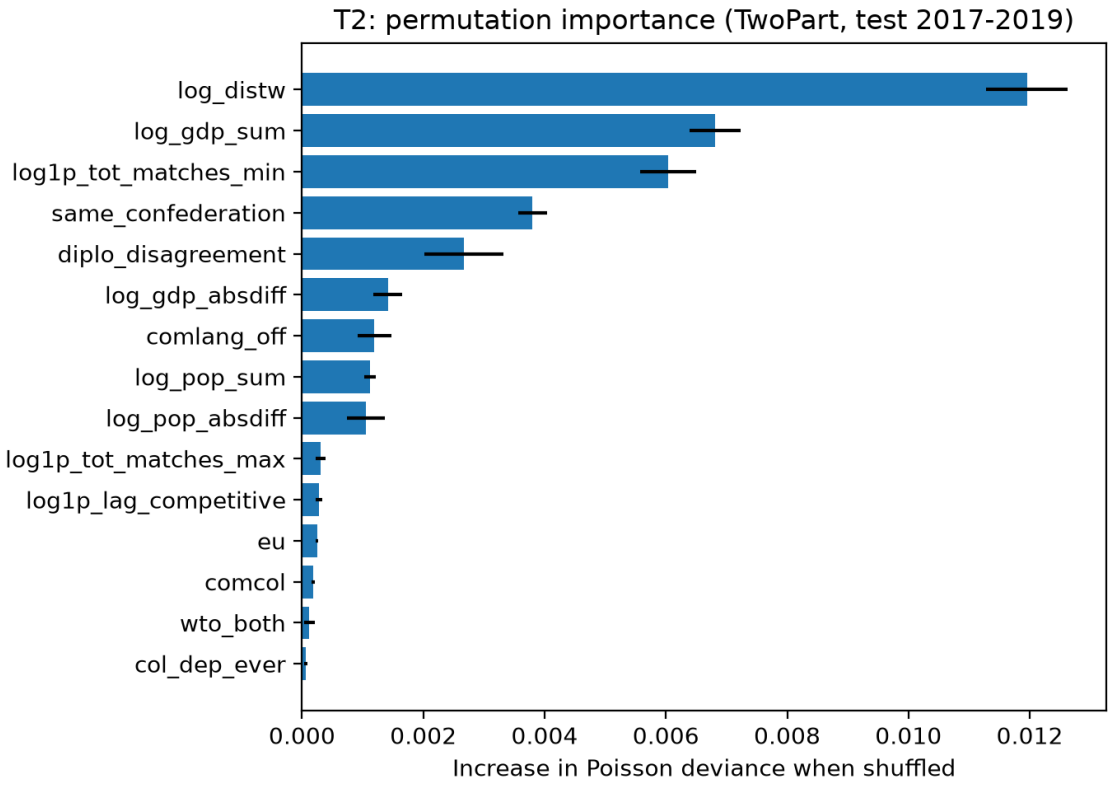
G+P+C vs G+P:

−0.2 to −0.5 points

Militarised disputes add no predictive power.

Performance

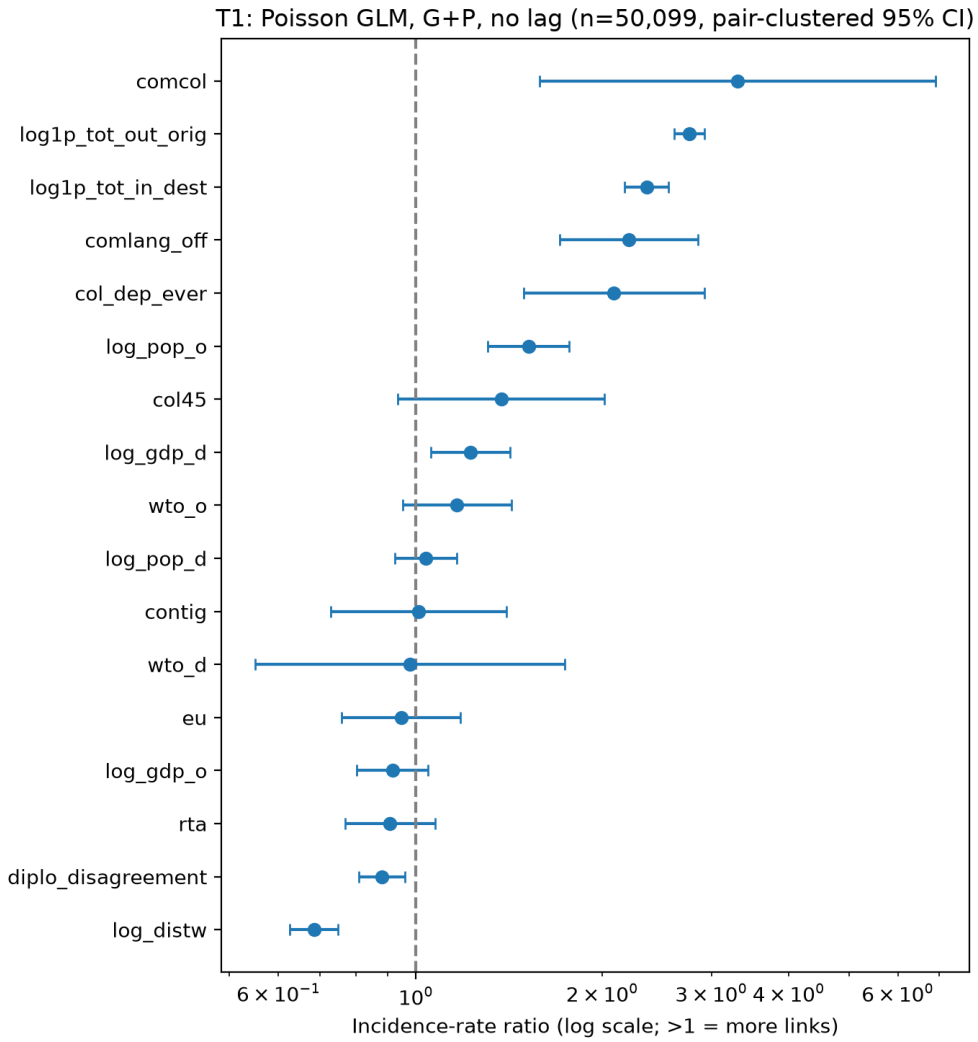




Feature Importance (Permutation, Test 2017–2019)

Transfers: market size and distance dominate. Friendlies: distance, economy, confederation; UN disagreement ranks 5th.

Performance



Talent Flows (T1): Effects (2010–2019)

Shared coloniser: × 3.3 more transfers

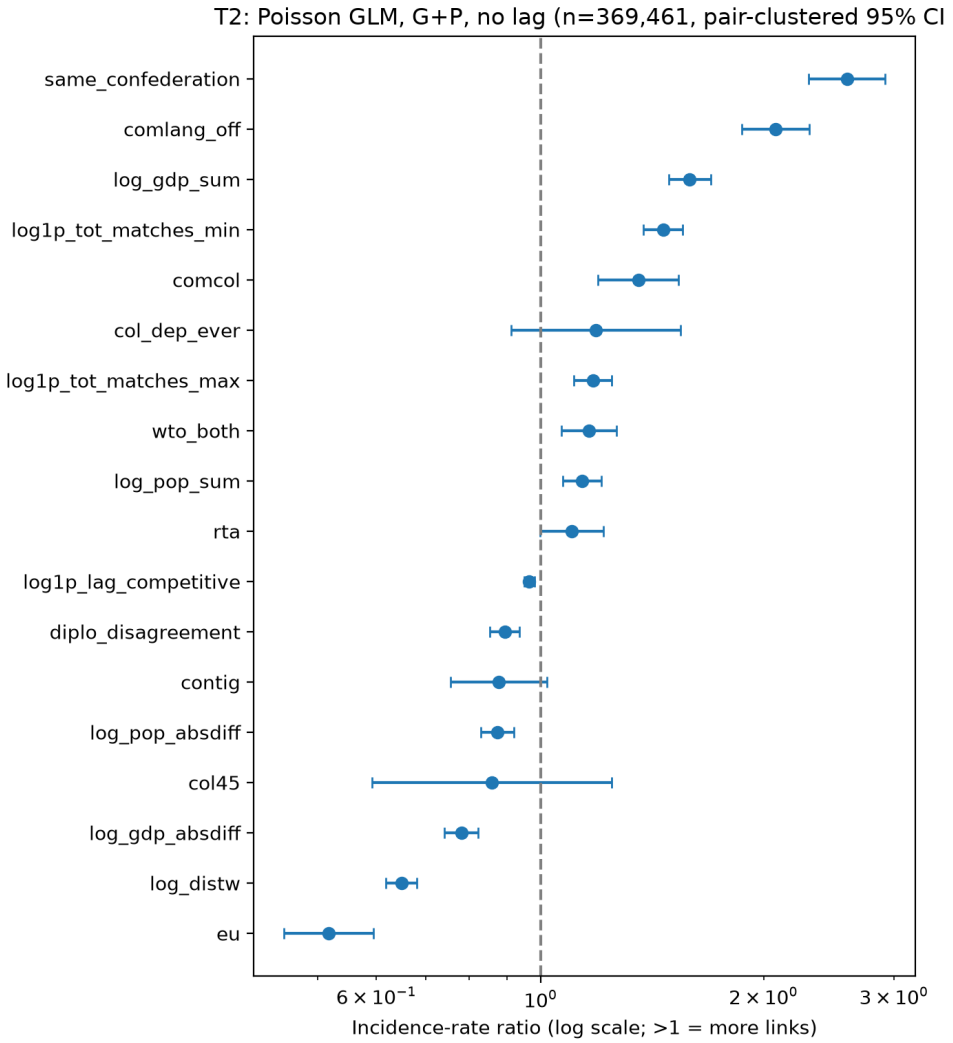
Ever colonial relationship: × 2.1

Common language: × 2.2

Distance (+1 SD): −32%

UN disagreement (+1 SD): −12%

Performance



Friendlies (T2): Effects (1994–2019)

Same confederation: × 2.6 more friendlies

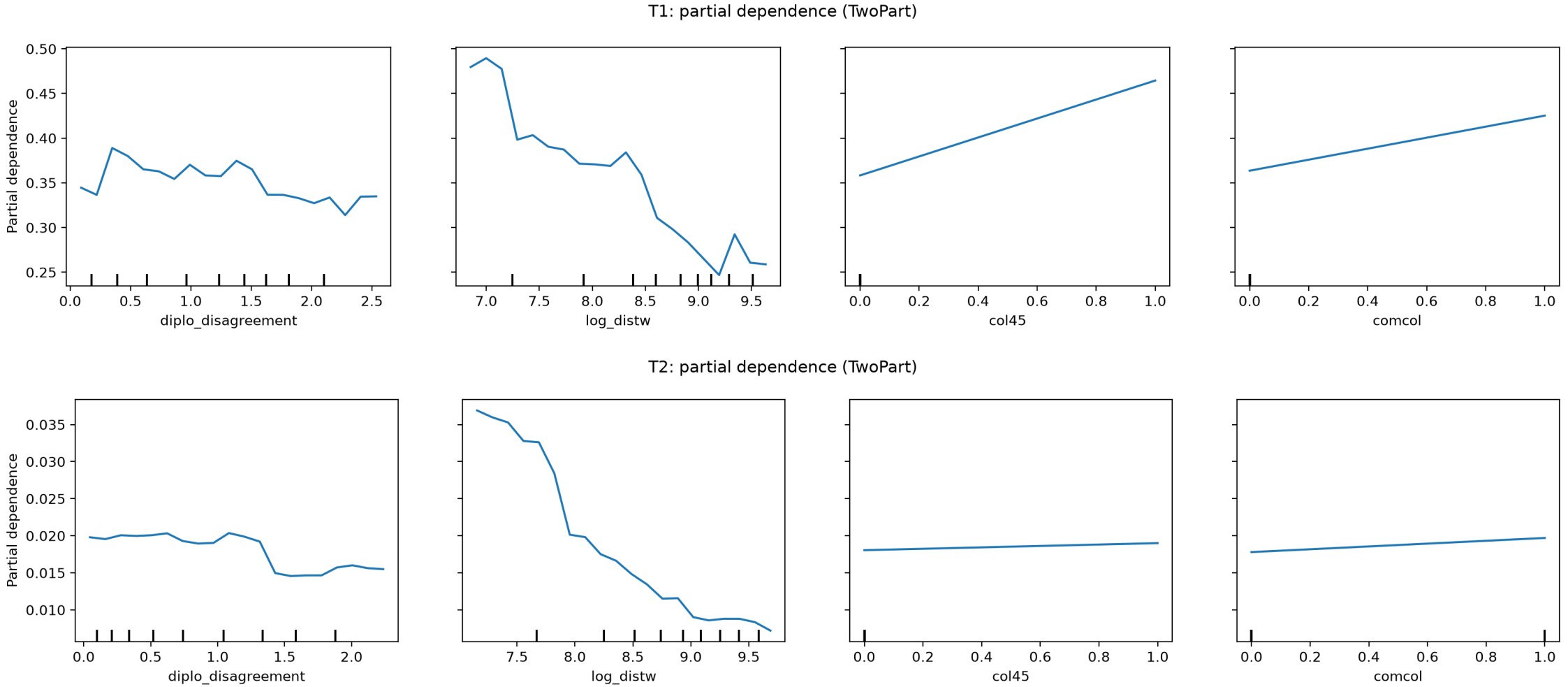
Common language: × 2.1

Distance (+1 SD): −35%

UN disagreement (+1 SD): −11%

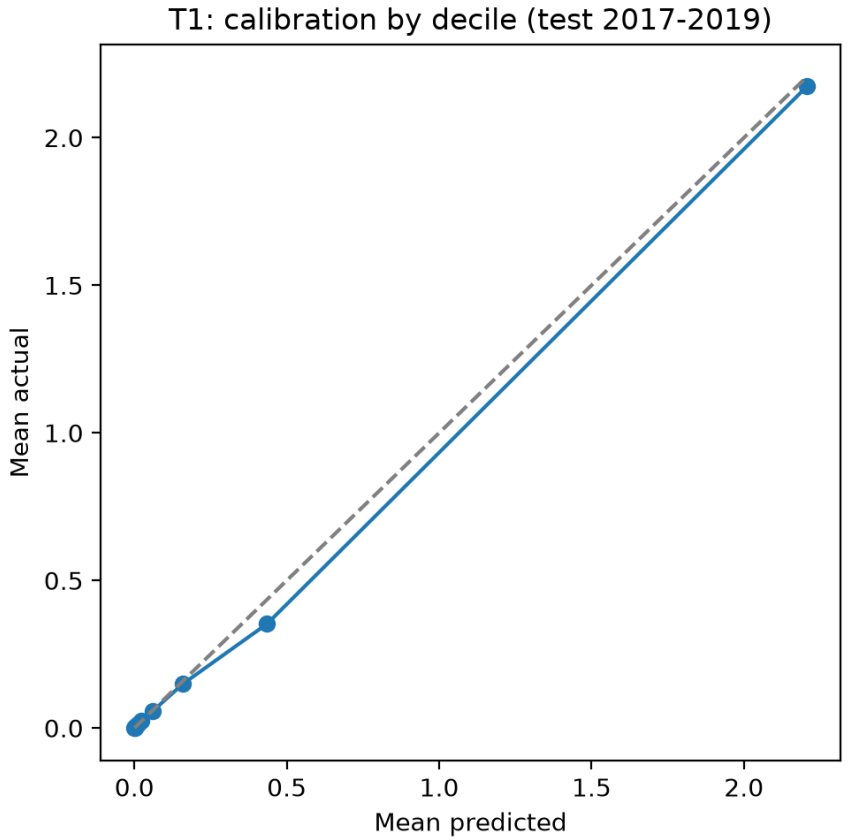
Both in the EU: −48% (crowded out)

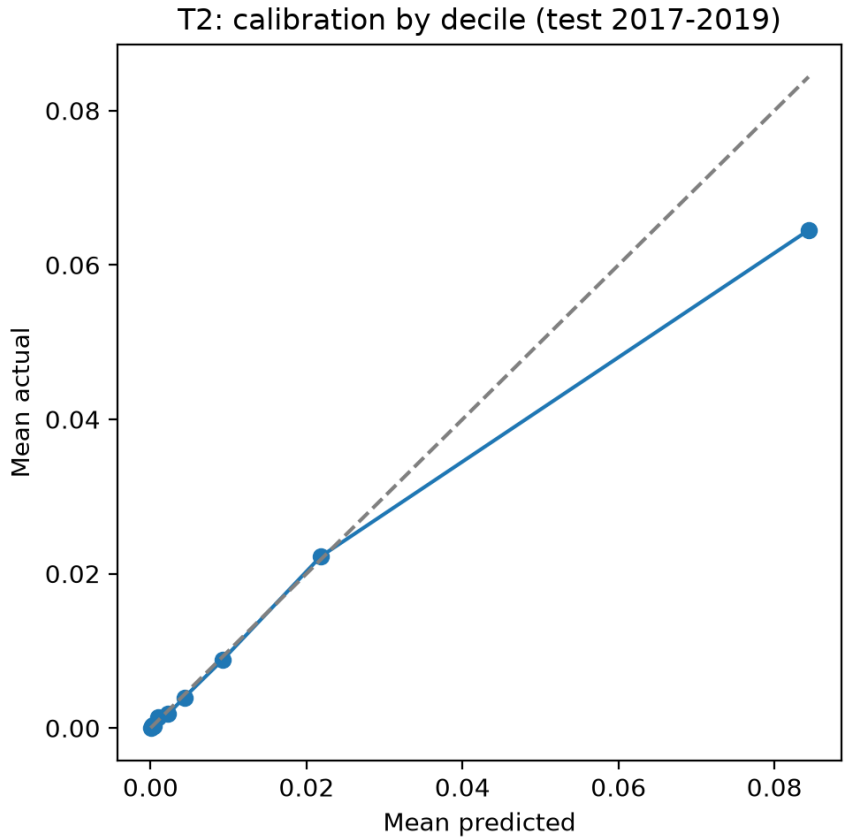
Performance



Partial Dependence: links fall with distance and with UN disagreement

Performance





Actual vs Predicted

Points on the dashed diagonal mean predicted amounts match reality; T2 slightly over-predicts the busiest pairs.

Key Finding: States React, Markets Don’t

Within-pair test: the same two countries compared with themselves over time (fixed-effects Poisson)

Football Link	Effect of +1 SD UN Disagreement	p-value	Placebo: Future Disagreement	Verdict
Friendlies (state choice)	−7.3%	0.04	−2.9% (n.s.)	Real but modest
Transfers (club market)	+9.9%	0.48	−2.3% (n.s.)	No effect

Why it matters

Friendlies are picked by federations and governments, and they **react to politics**. Transfers are picked by clubs, and they **do not**.

Honest caveat

Adding future disagreement shrinks the friendlies effect to −5% (p = 0.24): **suggestive, not proven**.

Key Finding: Football Mirrors Diplomacy, It Doesn't Make It

Reverse test: do more friendlies this year predict countries moving closer at the UN next year?

p = 0.32

Friendlies → next-year political change (17,564 pairs, 1994–2019)

p = 0.11

Player transfers → political change (2010–2019)

+0.005

Gain in R^2 when football features are added to an ML forecast of political change

No measurable effect: sport diplomacy **does not move UN voting** at this scale. A null result that challenges the popular "ping-pong diplomacy" story.

Case Study: Russia After the 2022 Invasion

Share of Russia's opponents whose country condemned the invasion (UN vote ES-11/1)

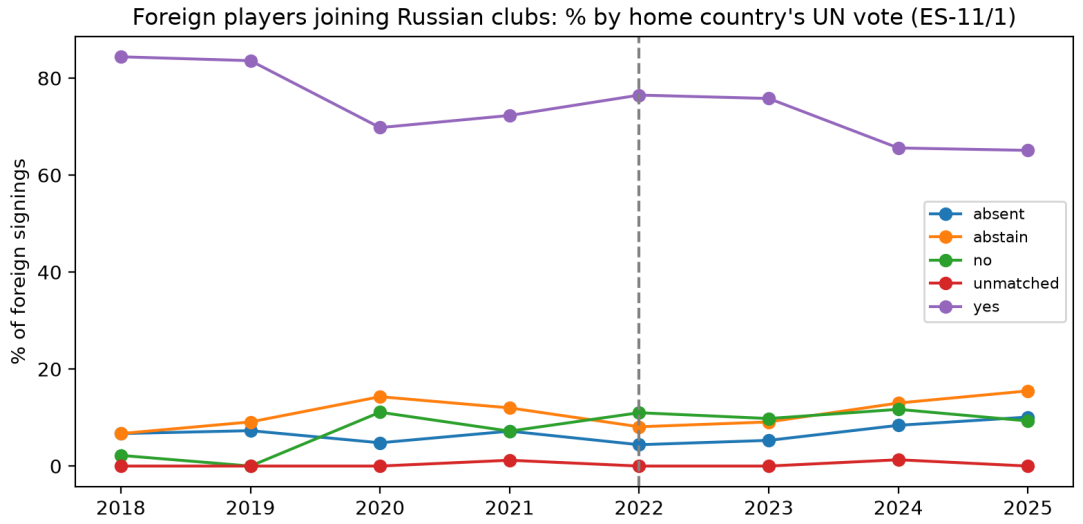
96%

Before invasion
46 of 48 matches

43%

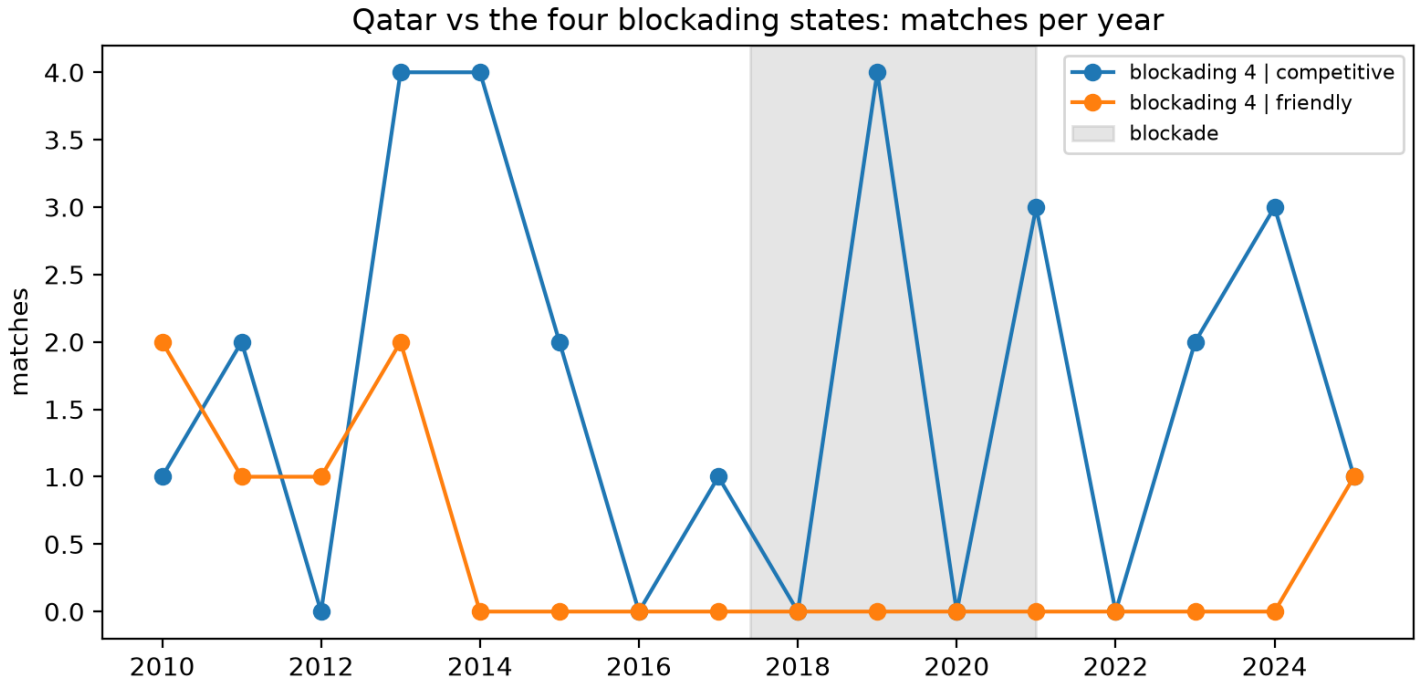
After invasion
10 of 23 matches

Post-invasion partners: Central Asia, Iran, Iraq, Cuba, Belarus, Syria. Only 27% of UN members did not condemn, yet they were **57% of Russia's opponents**.



Signings into Russian clubs: share from condemning countries fell from about 84% to 65%, but the slide began in 2020; Belarus became the 2nd-largest source.

Case Study: Qatar and the Gulf Crisis



What Happened

Friendlies with Saudi Arabia, UAE, Bahrain and Egypt **stopped in 2014**, the year of the first ambassador crisis, 3 years before the 2017 blockade.

Competitive matches (forced by tournaments) continued. The **chosen link broke**, the **obligatory link survived**. Small counts: a case study, not proof.

Limitations

Data Coverage: Transfermarkt cross-border records are thin before 2010; only 30 destination leagues; citizenship is current, not at transfer time.

Associations, Not Causes: within-pair and placebo tests strengthen the evidence but do not prove causality.

Conflict Data Ends in 2010: recent conflicts (e.g. Russia–Ukraine) are covered only by case studies.

Coarse Politics Measure: UN voting captures alignment, not day-to-day diplomacy; event case studies rest on small counts.

Conclusion

Summary: football ties follow gravity and history (distance, market size, language, colonial ties): transfers 72% deviance explained (AUC 0.93), friendlies AUC 0.90.

Geopolitics is associated but redundant for prediction: UN voting and colonial ties change accuracy by less than 0.5 points; conflict adds nothing.

Within the same pair, rising political tension is linked to ~7% fewer friendlies ($p = 0.04$) but not fewer transfers: states react, markets don't.

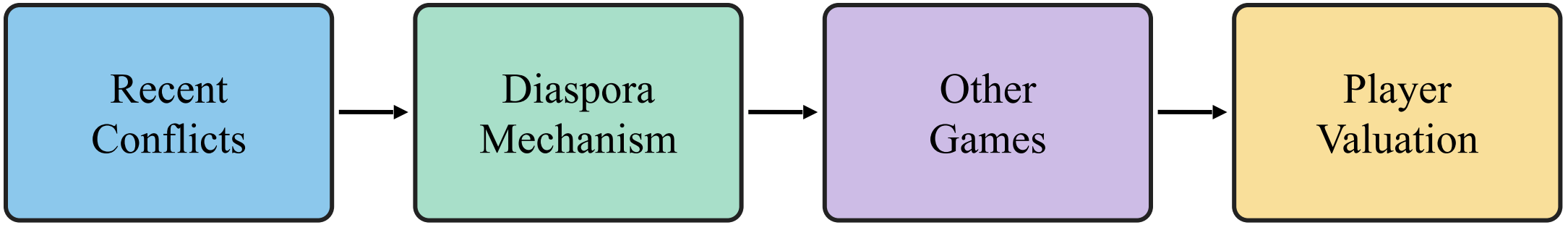
Final Verdict: football mirrors diplomacy rather than making it. Friendlies did not predict political rapprochement, while shocks (Qatar 2014, Russia 2022) reshaped who plays whom.

Learnings

1. **Data engineering is most of the work:** Country names, defunct states and coverage gaps decided the results
2. **Choose the right metric:** With 99% zeros, accuracy misleads; deviance, D^2 and AUC tell the truth
3. **Control leakage:** Lag every feature; split by country pair, not by row
4. **Prediction ≠ explanation ≠ causation:** Ablation, interpretation and fixed effects answer different questions
5. **A null result is still a result:** No effect is evidence too, when the test is sound

Future Work

- **Recent Conflicts:** extend beyond 2010 with UCDP conflict data and GDELT news tone
- **Diaspora Mechanism:** add UN migrant stock to test whether colonial effects work through migrant communities
- **Other Games:** compare women's football and cricket
- **Player Valuation:** are players from politically distant countries undervalued for their performance?



Related Work

S.No.	Content Type	Title of Content	Linking Content
1	RESEARCH PAPER	Leg Drain: Quantifying the Global Redistribution of Football Talent through Multi-National Eligibility	A. Lehner, G. Righetto (arXiv, 2026)
2	RESEARCH PAPER	The Diplomatic Role of Football in Contemporary Society	N. A. Nesterenko
3	RESEARCH PAPER	Sport and Diplomacy in International Relations	S. Chandel, A. P. Singh (IJSRST, 2025)
4	DATASET PAPER	The CEPII Gravity Database	M. Conte, P. Cotterlaz, T. Mayer (2022)
5	DATASET PAPER	An Analysis of the Militarized Interstate Dispute (MID) Dataset	D. M. Gibler, S. V. Miller, E. K. Little (2016)

References

[1] A. Lehner and G. Righetto, “Leg Drain: Quantifying the Global Redistribution of Football Talent through Multi-National Eligibility,” arXiv:2603.17336, Mar. 2026.

[2] N. A. Nesterenko, “The Diplomatic Role of Football in Contemporary Society,” Ukrainian State University of Science and Technologies, Dnipro, Ukraine.

[3] S. Chandel and A. P. Singh, “Sport and Diplomacy in International Relations,” Int. J. Sci. Res. Sci. Technol., vol. 12, no. 6, pp. 769–783, Nov.–Dec. 2025.

[4] M. Conte, P. Cotterlaz, and T. Mayer, “The CEPII Gravity Database,” CEPII Working Paper no. 2022-05, 2022.

[5] D. M. Gibler, S. V. Miller, and E. K. Little, “An Analysis of the Militarized Interstate Dispute (MID) Dataset, 1816–2001,” Int. Stud. Q., vol. 60, no. 4, pp. 719–730, 2016.

[6] D. Cariboo, “Football Data from Transfermarkt,” GitHub: dcaribou/transfermarkt-datasets. [Online].

[7] M. Jürisoo, “International Football Results from 1872,” GitHub: martj42/international_results. [Online].

[8] V. Arel-Bundock, N. Enevoldsen, and C. J. Yetman, “countrycode: An R package to convert country names and country codes,” J. Open Source Softw., vol. 3, no. 28, p. 848, 2018.

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[10] J. M. C. Santos Silva and S. Tenreyro, “The Log of Gravity,” Rev. Econ. Stat., vol. 88, no. 4, pp. 641–658, 2006.

[11] United Nations General Assembly, Resolution ES-11/1, “Aggression against Ukraine,” 2 Mar. 2022.